

Configuring Telnet Remote Access in Packet Tracer

Standard Operating Procedure & Implementation Guide for Cisco Routers (/24 Subnet)

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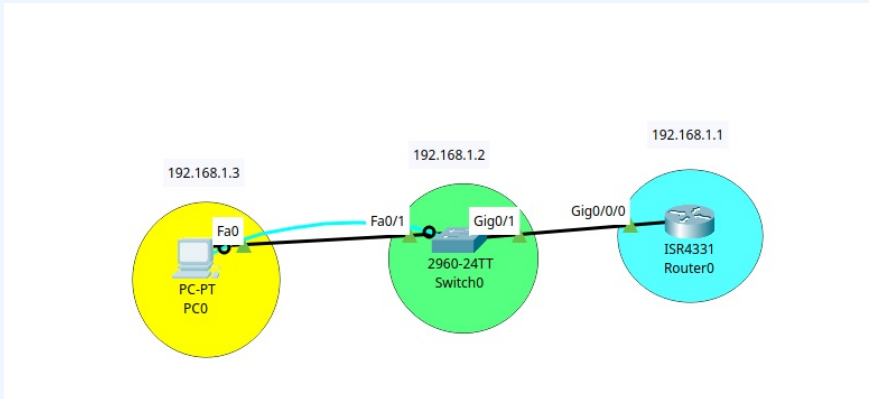
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Environment: Cisco Packet Tracer

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1. Network Topology & Overview

This documentation outlines the step-by-step procedure to enable remote management (**Telnet**) on a Cisco Router. The network uses a /24 prefix (Subnet Mask: 255.255.255.0). PC0 connects to the router through a Layer 2 Switch and initiates remote management via the Packet Tracer Desktop Telnet GUI tool.



2. IP Addressing Scheme (/24 Subnet)

DEVICE NAME	INTERFACE	IP ADDRESS	SUBNET MASK	DEFAULT GATEWAY	ROLE
Router0	GigabitEthernet 0/0/0	192.168.1.1	255.255.255.0 (/24)	N/A	Default Gateway & Telnet Server
Switch0	Fa0/1, Gig0/1	192.168.1.2	255.255.255.0 (/24)	192.168.1.1	Layer 2 Switch (Management SVI)
PC0	FastEthernet 0	192.168.1.3	255.255.255.0 (/24)	192.168.1.1	Telnet Client Workstation

3. Router Configuration Steps (Cisco IOS)

Step 1 Configure Default Gateway LAN Interface

Assign IP 192.168.1.1 with subnet mask 255.255.255.0 to the GigabitEthernet interface.

```
Router0> enable
Router0# configure terminal
Router0(config)# interface GigabitEthernet 0/0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
```

Step 2 Define Hostname

Set the hostname to Router0 for identification within the topology.

```
Router0(config)# hostname Router0
```

Step 3 Set the Privileged EXEC (Enable) Password

Restrict access to privileged EXEC mode with an encrypted secret.

```
Router0(config)# enable secret class
```

Step 4 Configure Virtual Terminal (VTY) Lines for Telnet

Restrict incoming remote access on virtual lines 0 through 4 to Telnet only, and set the session password.

```
! Configure the VTY password and enforce Telnet
Router0(config)# line vty 0 4
Router0(config-line)# password cisco
Router0(config-line)# login
Router0(config-line)# transport input telnet
Router0(config-line)# exec-timeout 5 0
Router0(config-line)# exit
```

Step 5 Save the Running Configuration

Commit the configuration to NVRAM so it persists across a reboot.

```
Router0(config)# end
Router0# copy running-config startup-config
```

4. Testing Remote Access via Packet Tracer Desktop Telnet App

A Connecting via Desktop Telnet Client GUI

1. Click on PC0 (IP: 192.168.1.3, Mask: 255.255.255.0) in Packet Tracer and navigate to the **Desktop** tab.
2. Open the **Command Prompt** application.
3. Enter the command: `telnet 192.168.1.1`
4. Enter the VTY password `cisco` when prompted.
5. Enter `enable` and the secret `class` to reach privileged EXEC mode.

B Verify Active Session on Router

Run operational commands on Router0 to confirm active Telnet sessions:

```
Router0# show ip interface brief
Router0# show line
Router0# show users
```

Security Note (Telnet vs. SSH)

Telnet transmits all data — including the login password — in plaintext over TCP port 23. It is suitable for isolated lab environments only. For production or internet-facing devices, use SSH (see NET-DOC-SSH-001) instead.